CS2103 / Lab – Assignment 10

Topic: Information Retrieval – Vector Space Model and TF–IDF

Instructions: Complete all five questions. Implement the **document representation, similarity measures, and TF–IDF calculations from scratch using Python/NumPy**. Built-in vectorizers, TF–IDF, or similarity functions are not allowed except for verification.

Q1. Bag of Words and Term–Document Matrix

Use the following documents from the lecture:

$$d_1 = \text{“The cow jumped over the moon”},$$
$$d_2 = \text{“O’Leary’s cow kicked the lamp”},$$
$$d_3 = \text{“The kicked lamp started a fire”},$$
$$d_4 = \text{“The cow on fire”}.$$

Build the vocabulary automatically and represent every document using **Bag of Words**. Construct and display the complete **Term–Document Matrix** containing the frequency of every term in every document.

Q2. Euclidean Distance and Document Length

Implement Euclidean distance between two document vectors,

$$dist(X, Y) = \sqrt{\sum_{i=1}^{k} (x_i - y_i)^2}.$$

Compute the pairwise distances among the four documents and identify the nearest document pair.

Then compare documents of different lengths and briefly explain why **document length can affect Euclidean-distance-based similarity**.

Q3. Cosine Similarity from Scratch

Implement cosine similarity,

$$sim(X, Y) = \frac{X \cdot Y}{\|X\| \, \|Y\|}.$$

Compute all pairwise cosine similarities and rank d_1, d_2, d_3 according to their similarity with d_4.

Compare this ranking with the Euclidean-distance result from Q2 and explain why cosine similarity is useful for document comparison.

Q4. TF–IDF Term Weighting

For each term i in document j, compute the TF–IDF weight using

$$w_{i,j} = \big(1 + \log(tf_{i,j})\big) \log \frac{N}{df_i},$$

where $tf_{i,j}$ is the term frequency, df_i is the number of documents containing term i, and N is the total number of documents.

Construct and display the complete **TF–IDF matrix**. Identify terms receiving high and low weights and explain why common terms are assigned smaller weights.

Q5. Retrieval Similarity Challenge

Using the TF–IDF vectors obtained in Q4, recompute

$$sim(d_1, d_4), \qquad sim(d_2, d_4), \qquad sim(d_3, d_4).$$

Rank d_1, d_2, d_3 according to their similarity with d_4 using:

(i) raw term-frequency vectors,
(ii) TF–IDF vectors.

Compare the two rankings and explain how TF–IDF changes the influence of common and informative terms on document similarity.

Note: Students must be able to explain Bag of Words, Vector Space Model, term frequency, Term–Document Matrix, Euclidean distance, cosine similarity, document length effects, document frequency, and TF–IDF during the evaluation/viva.